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On the causality and plausibility of treatment effects in operations management research

Sunil Mithas[1] | **Yanzhen Chen**[2] | **Yatang Lin**[3] | **Alysson De Oliveira Silveira**[1]

[1]School of Information Systems and Management, University of South Florida, Tampa, Florida, USA

[2]Department of Information Systems, Business Statistics and Operations Management, Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong

[3]Department of Economics, Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong

Correspondence
Sunil Mithas, University of South Florida, Tampa, FL, USA.
Email: smithas@usf.edu

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Abstract

Empirical research in operations management (OM) has made rapid strides in the last 30 years, and increasingly, OM researchers are leveraging methods used in the econometrics and statistics literature to assess the causal effects of interventions. We discuss the two key challenges in assessing causality with observational data (i.e., baseline bias, differential treatment effect bias) and how dominant identification approaches such as matching, instrumental variables, regression discontinuity, difference-in-differences, and fixed effects deal with such challenges. We surface the key underlying assumptions of different causal estimation methods and discuss how OM scholars have used these methods in the last few years. We hope that reflecting on the plausibility and substantive meaning of underlying assumptions regarding different identification strategies in a particular context will lead to a better conceptualization, execution, evaluation, dissemination, and consumption of OM research. We conclude with a few thoughts that authors and reviewers may find helpful in their research as they engage in discourse related to causality.

KEYWORDS

baseline bias, causality, differential treatment effect bias, empirical research, endogeneity, observational studies, quasi-experiment, selection

1 | INTRODUCTION

As operations management (OM) research becomes increasingly more empirical over time (Gupta et al., 2006; Melnyk et al., 2018; Roth & Rosenzweig, 2020; Terwiesch et al., 2020), causality and endogeneity issues are receiving far more attention among OM scholars than was the case earlier (Ho et al., 2017; Ketokivi & McIntosh, 2017; Lu et al., 2018). The focus on assessing causality in empirical OM research is consistent with the interest in understanding the causal effect of interventions in other disciplines (Gow et al., 2016; Mithas & Krishnan, 2009; Mithas et al., 2006, 2014; Mithas, Xue, et al., 2022; Morgan & Winship, 2007; Roberts & Whited, 2013; Rosenbaum, 2019). Indeed, the alphabetical soup of propensity score matching (PSM), regression discontinuity (RD), instrumental variable (IV), difference-in-differences (DID), and fixed effects (FEs) is now witnessed far more frequently in OM articles, compared to just a decade ago.

Our objective in this paper is to describe how OM research has so far used econometric and counterfactual (potential outcomes) approaches for assessing causality, focusing particularly on papers published in *Production and Operations Management* (*POM*). We present the key challenges in assessing causality using a potential outcome framework and discuss some of the dominant identification strategies. We begin by reviewing empirical *POM* papers published during 2016–2021 to identify the key approaches used by OM scholars for causal inference when using observational data. Subsequently, we discuss two critical challenges that various identification strategies address. We discuss the advantages and challenges of implementing different identification strategies and discuss some *POM* articles as examples. We conclude with a discussion and a few observations.

Our review of different identification strategies in the *POM* empirical papers for 2016–2021 reveals interesting patterns and complements prior work that reviewed causality in OM papers for 2010–2015 (Ho et al., 2017).[1] Following the approach used by Gupta et al. (2006), we identified 411 articles published in *POM* from 2016 to 2021 that we classified

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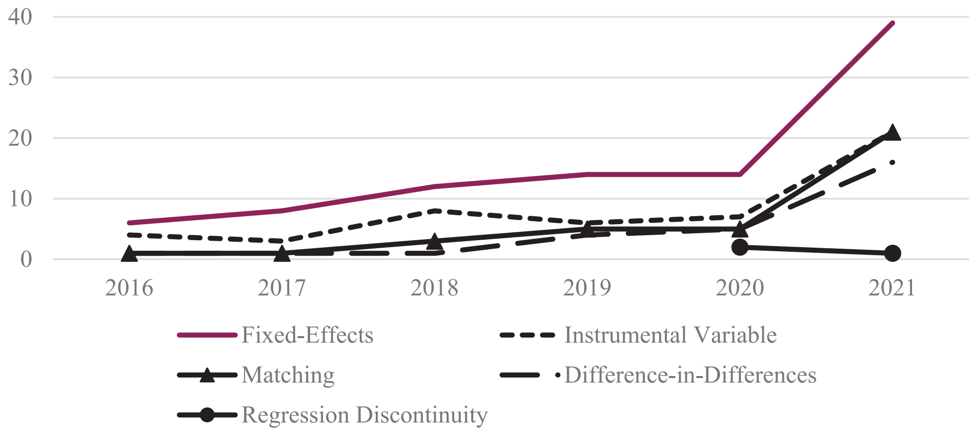

FIGURE 1 Yearly frequency of use of common identification strategies in *Production and Operations Management* (*POM*) [Color figure can be viewed at wileyonlinelibrary.com]

as meeting the definition of empirical papers. Among the 197 (48% of empirical papers) papers that employed either associational/causal techniques (the remaining 214 [52%] papers used analytical/predictive models), 122 papers used at least one of the following identification approaches commonly applied in the counterfactual framework: matching, IVs, RD, DID, or FEs.

Figure 1 shows the frequency of papers adopting each identification strategy over the years. Some papers combine several identification strategies.[2]

2 | KEY CHALLENGES IN ASSESSING CAUSALITY IN OBSERVATIONAL STUDIES AND TRADITIONAL REGRESSION

To clarify the critical challenges in assessing causality, we use the notation related to the potential outcomes—or counterfactual—approach and discuss how it illuminates the fundamental problem of causal inference and various types of causal effects that one can estimate based on the underlying assumptions and availability of data.[3] This discussion also sheds light on two critical challenges in assessing causality: *baseline bias* and *differential treatment effect bias*. We then relate the potential outcomes approach to the most familiar regression framework that OM researchers frequently use in their empirical work.

2.1 | Potential outcomes or counterfactual framework

Consider a population Q with its individuals or units denoted by i, where lowercase z represents the treatment, which can take at least two distinct values or levels. Typically, $z = 0$ represents the standard, baseline, low, or no treatment. For every unit i, and corresponding to each value of treatment z, there is a potential response (or outcome) $Y_i(z)$, which may be continuous or discrete. For example, consider studying the causal effect of investing in enterprise resource planning (ERP) systems on *better information sharing* among the population of Fortune 500 firms. In this case, z is dichotomous, depending on whether a firm has implemented ERP ($z = 1$, the treatment) or not ($z = 0$, the control), and one can conceptualize two potential responses for the benefit from ERP in terms of *better information sharing*: $Y_i(0)$ and $Y_i(1)$.

2.1.1 | Variety of treatment effects: average treatment effect (ATE), average treatment effect among treated (ATT), and average treatment effect among untreated (ATU)

With this notation, one can define the ATE in the population Q as follows:

$$ATE = E(Y(1) - Y(0)), \tag{1}$$

where the expectation operates on the potential outcomes across all units in the population of interest Q (note that the subscript i has been omitted).

The main challenge of causal identification arises because instead of observing $E(Y(1))$ and $E(Y(0))$, we observe $E(Y(1)|Z = 1)$ and $E(Y(0)|Z = 0)$, and we end up estimating a *prima facie standard estimator denoted by* T in Equation (2), and rewriting T by simply adding and subtracting $E(Y(0)|Z = 1)$ in Equation (3):

$$T = E(Y(1)|Z = 1) - E(Y(0)|Z = 0), \tag{2}$$

$$T = \left[E(Y(1)|Z = 1) - E(Y(0)|Z = 1)\right] + \left[E(Y(0)|Z = 1) - E(Y(0)|Z = 0)\right]. \tag{3}$$

The first term in Equation (3) is a conditional ATE called the ATT. The second term $[E(Y(0)|Z = 1)$

$- E(Y(0)|Z=0)]$ describes the difference between the potential "counterfactual" outcomes for the treated units if they had not been treated and the actual outcomes for the control units.

Note that ATE is a weighted average of ATT and the ATU, with π being the proportion of treated units (Morgan & Winship, 2007), such that

$$ATE = \pi * ATT + (1-\pi) * ATU. \tag{4}$$

We can also rewrite T as

$$T = ATE + \left[E(Y(0)|Z=1) - E(Y(0)|Z=0)\right] + (1-\pi) * [(ATT - ATU)]. \tag{5}$$

In observational studies, T may reflect a combination of the true ATE plus another quantity that reflects a non-causal association between Z and the potential outcomes $Y(0)$ and $Y(1)$, which is the sum of the second and third terms. The second term $[E(Y(0)|Z = 1) - E(Y(0)|Z = 0)]$ represents the *baseline bias*, while the third term $(1-\pi) *[(\text{ATT} - \text{ATU})]$ reflects the *differential treatment effect bias* (Winship & Morgan, 1999; Xie, 2011).[4]

2.1.2 | Two challenges in assessing causality

We elaborate the baseline bias and differential treatment effect bias as two separate issues. Baseline bias arises when the counterfactual outcomes for the treated unit in the untreated state differ from the observed baseline outcomes for the control units. For example, if we divide all firms into those that have adopted ERP systems and those that have not, in a hypothetical world where the treatment had never happened to any of the groups, the "baseline outcomes" should be, on average, the same between these two groups. However, in observational studies, because the treatment assignment status is typically not random, this condition does not hold. Two important reasons are (1) *omitted variable bias* and (2) *simultaneity*. In the case of omitted variable bias, some factors affect the selection of the treatment and the outcome that are unknown or unavailable to researchers. Another reason for baseline bias can be simultaneity, which occurs when the dependent and independent variables affect each other. In other words, X causes Y, but Y also causes X.[5]

Besides the baseline bias, differential treatment effect bias may also render our "naïve" or standard estimate of T biased in observational data if $ATT \neq ATU$. For example, it is likely that those who decided to be in the treatment were more likely to benefit from it, even if one assumes that there were no baseline differences in the absence of the treatment. For instance, when examining the impacts of ERP adoption on firm outcomes, adopters are likely more prone to benefit from the implementation than those who decided not to adopt this system (e.g., $ATT > ATU$). Note that treatment heterogeneity is a conceptually separate issue from baseline bias, which relates to differences in the potential outcomes in the untreated state for both the treated and control units. However, the differential treatment effect suggests that even if we could rule out any baseline differences and estimate the treatment effects on the treated units (ATT), such an estimate might not be generalizable to those in the control group. This discussion suggests that if one does not want to assume a constant treatment effect (as is usual in many regression-based and widely used econometric methods) for estimating ATE, it may be possible to estimate ATT by ignoring the differential treatment effect bias if one focuses on a narrower research question involving ATT.

Note that randomization in experimental studies addresses *both* baseline bias and differential treatment effect bias by making both the unobserved factors and potential outcomes independent of the treatment status enabling the estimation of ATE. However, the lack of randomization in observational studies suggests that alternative empirical methods most likely can only estimate ATT or even narrower treatment effects, such as the local ATE (LATE).

2.2 | Traditional regression approach

The potential outcomes approach clarifies the assumptions of the traditional regression-based approach that OM researchers widely use. The traditional regression framework posits models for the observed response Y given observed treatment Z and observed covariates X.[6] Assuming that the response Y is continuous, Z is binary, and X is a vector of discrete or continuous covariates, regression models typically take the form where an ordinary least squares estimate of β is typically said to describe an estimate of the treatment effect of Z on Y.

$$Y_i = \alpha + \beta Z_i + X_i\gamma + U_i. \tag{6}$$

When does β have a valid interpretation as the causal effect of the treatment on the response if we condition (i.e., "adjust") on the observed confounders X as in Equation (6)? The language of the potential outcomes clarifies this issue (Sobel, 1995) by suggesting that the model in Equation (6) *implies* that β is the difference in the conditional expectations of the response given X for $Z = 1$ versus $Z = 0$:

$$\beta = E(Y|Z=1, X=x) - E(Y|Z=0, X=\text{x}). \tag{7}$$

Note that Equation (7) does not represent a contrast between $Y(0)$ and $Y(1)$; therefore, within the potential outcomes framework, it does not represent a causal effect or the average causal effect of the binary treatment z on $Y(z)$: $ATE = E(Y(1) - Y(0))$.

Interpreting β causally requires making additional assumptions about the relationship between the observed data X, Z, Y, and the potential outcomes $Y(0)$, $Y(1)$. In particular, we will need to assume that the selection into treatment is entirely driven by Xs, and there are no further unobserved factors that jointly determine both Y and Z as follows:

$$Y(0),\ Y(1)\ \perp Z \text{ given } X,\ \textit{or } U \perp Z \text{ given } X, \tag{8}$$

where the symbol ⊥ denotes independence. This condition is known as the conditional independence assumption (CIA), also referred to as the "strong ignorability assumption" in the statistics literature (Rosenbaum & Rubin, 1983b).[7] Only when we assume conditional independence or strong ignorability, then β from the regression model in Equation (6) is equivalent to ATE (Pratt & Schlaifer, 1988).[8] Regression-based models are familiar and easy to execute, and they are fine to use for descriptive purposes. However, there is a need for caution in the use of all-cause kitchen sink models for causal inference in the presence of weak theories concerning the functional form of all covariates, and when one suspects significant treatment effect heterogeneity (Morgan & Winship, 2007). Indeed, Achen (2002) cautions that regression can provide wrong estimates for a variable if some other variable is entered with an incorrect functional form even when one is dealing with just two covariates, it is much more difficult to explore the implications of such non-linearities when one is dealing with multiple covariates.

3 | IDENTIFICATION STRATEGIES FOR ASSESSING CAUSALITY IN *POM* EMPIRICAL PAPERS

We next discuss several identification strategies: Matching (including PSM, coarsened exact matching (CEM), and entropy balancing [EB]); the IV approach; RD; FEs, the DID, and synthetic control (SC) approaches. The nature of the questions and the types of data often determine the choice of one or more approaches. We first describe methods for cross-sectional data, followed by methods primarily applied to panel data.

3.1 | Approaches based on matching

A critique of controlling covariates directly in regressions is that the confounders might not be linear in the set of covariates. One alternative to address the issue with covariates is the matching method, which consists of selecting a control sample that is very similar to the treated units. The new pruned sample created by matching, in which the observed covariates of the treated and control units are very close, can subsequently be used in the regression. The matching method relies on the strong ignorability assumption:[9]

$$Y(0),\ Y(1) \perp Z \text{ given } X. \tag{9}$$

The matching method also uses another common support assumption which ensures that there is sufficient overlap in the characteristics of the treated and untreated units to find adequate matches:

$$0 < Pr(\, Z_i = \ 1|X_i) < 1 \text{ for all feasible } X_i. \tag{10}$$

The PSM method is the most popular among matching models. This popularity is partly because sample sizes are often not big enough to achieve perfect matching on all observed covariates. Rosenbaum and Rubin (1983b) propose a solution to the sample size problem based on a *propensity score*, which is the probability of being treated given the covariates among all treated and control units, usually estimated by a logit or probit model. The traditional PSM method usually prunes the sample by matching each treated subject with a single or a few control subjects based on the estimated probabilities of selection into the treatment. There are multiple algorithms from which to choose, such as nearest-neighbor matching and caliper matching. Compared to conventional regression analysis, some advantages of the propensity score approach are: (i) reduced dimensionality of the problem of matching on multiple covariates, (ii) better visibility of the extent to which the treatment and control groups are similar, and (iii) avoiding reliance on strong functional form assumptions (linear effect of covariates) that are implicit in the regression analysis (Dehejia & Wahba, 1999; Rubin, 1997).

However, the propensity score approach must also be used with caution, as it is a necessary but not sufficient condition to ensure the balance of the covariates.[10] Therefore, it is crucial to check the balance of the covariates, possibly through t-tests or plotting the treated versus control distributions after PSM. Another recommendation is to use the propensity scores as weights and not conduct any pruning. An alternative to PSM is CEM, which involves coarsening each covariate beforehand and applying exact matching to the coarsened X.[11] Although less commonly used than PSM, CEM is usually faster to implement and ensures an *ex-ante* known level of balance in the covariates. A potential downside of CEM is that it requires a large sample to find matches for treated units when the number of covariates is large.

EB is another method for matching treatment and control observations (Hainmueller, 2012) that uses a weighting approach. EB consists of assigning weights to observations to force certain balancing conditions, such as the approximate equalization of means, variances, and skewness between the treated and control units. Among its advantages, EB does not require researchers to make many decisions (e.g., checking the balance, choosing a propensity score cutoff, or picking the number of matched units to improve its quality); it only needs a list of covariates and the number of targeted moments. Additionally, EB does not require many observations to perform effective matching such as CEM and also does not involve dropping many observations from the sample, as the control units that look very different from the treated ones are assigned a small weight. However, EB would not produce a clearly defined "matched sample" through pruning, making it more difficult to compare the *distribution* of the covariates between the treated and matched samples. In addition, we can only ensure the balance in EB in targeted moments but not higher-level moments.

Among applications of matching identification strategies in *POM*, Ba et al. (2021) use PSM to study how adopting a mobile app changes consumer behavior with respect to a hotel chain. Because the treatment (mobile app adoption) was not randomized, those who chose to use the app may differ from those who did not. For instance, the adopters may have expected to benefit more from the app, or adopters may be younger individuals with different consumer behaviors than non-adopters, even in the absence of the app (baseline bias). Ba et al. (2021) use PSM with observables such as gender, third-party account association, availability of private information, nationality, membership level, and the amount of money users spend. Ba et al. (2021) also compare the distribution of the covariates before and after matching, including *t*-tests and Kolmogorov–Smirnov tests.[12] Mejia et al. (2019) use a CEM-based matching technique to study the impact of transparency on donations to crowdfunding campaigns. They compare campaigns that release updates with those that do not and discuss the use of CEM (using campaign characteristics such as financial goals and the number of images and videos) as a potential solution for the "omitted variable bias" (p. 1781) due to a lack of data on the intrinsic value of the organizer.[13] We only identified one recent study in *POM* that employs EB as a matching strategy (e.g., Chang et al., 2022) reported in a footnote in one sentence as a robustness check to PSM.

3.2 | Instrumental Variable (IV) approaches

Note that both regression-based approaches that use conditioning to adjust for other causes of outcome *Y* and matching approaches that use conditioning to balance the determinants of the causal intervention *T* only address selection due to observables. They are of little help when one suspects the selection on unobservables.

An IV approach can help solve problems due to the selection on unobservables, which arises when a regressor (in our case, a treatment) correlates with the disturbance term (Greene, 2000; Heckman, 1997). Suppose we want to identify the treatment effect of treatment *T* on *Y*, but we suspect that *T* correlates with the unobserved omitted variable *U*. A possible solution is to find another variable *Z*, an instrument that correlates strongly with *T* but not with *U*. In other words, the only channel by which *Z* might affect *Y* is through its impact on *T*. A critical application of the IV approach is to address the simultaneity problem: When causality runs both ways, *Z* helps identify the causal channel from *X* to *Y* and not the other way around.

Equation (11) shows the first stage of the IV approach in which it creates a predicted value of the treatment variable *T*. Assuming that $cov\ (Z_i, u_i) = 0$, then $\hat{T}$ contains variation in T_i that is uncorrelated with u_i in the structural equation (for the causal effect of *T*).

$$\hat{T}_i = \widehat{\beta}_0 + \widehat{\beta}_1 Z_i, \tag{11}$$

$$Y_i = \alpha_0 + \alpha_1 \widehat{T}_i + u_i. \tag{12}$$

The traditional IV estimator with constant treatment effects assumption must meet two conditions: relevance and exclusion restriction. Relevance implies that the IV *Z* and the treatment are correlated. A weak correlation can create a weak IV problem (Stock et al., 2002); as a rule of thumb, the *F*-statistic of a joint significance test when all excluded instruments (the variables in *Z* that are not in *T*) are significantly different from zero should be bigger than 10 in the case of a single endogenous regressor. The second condition (exclusion restriction) requires that the effect of the proposed instrument on the outcome exclusively operates through its effect via the treatment, that is, Cov(Zi,ui) = 0. We can never statistically test it because it involves unobserved omitted variables U. As a result, it is often hard to find suitable instruments (Bound et al., 1995; Cascio & Lewis, 2006). Therefore, instead of finding perfect instruments, it may be more appropriate to establish the plausibility of the chosen instruments (Peng & Lai, 2011).[14]

The IV estimator can also accommodate heterogeneous treatment effects in the potential outcomes approach and clarifies that the IV approach identifies the treatment effects on only those units whose treatment status will change in response to the IV. In other words, even if an exclusion restriction and an additional monotonicity condition (the instrument either leaves the treatment unchanged or affects it in one direction only) for the treatment assignment are satisfied, the IV estimator is an estimate of the LATE for compliers in a population (Angrist et al., 1996), and the LATE estimate becomes specific to the choice of a particular instrument.[15] In other words, the use of IV is not a panacea by itself; in some cases (Crown et al., 2011), a better choice may be *not* to use an IV approach if the researcher cares about treatment effects on a broader group.

In sum, the potential outcomes approach provides new ways of thinking about the benefits and pitfalls of the IV approach and the potential to define and interpret narrower causal effects in the face of the heterogeneity of treatment effects such as LATE, marginal treatment effects (MTE), and the use of monotone IVs for the analysis of Manski's bounds (Morgan & Winship, 2007).

The IV approach is one of the most common identification strategies in *POM* papers published since 2016. Among examples of using lagged independent variables as instruments, Tang et al. (2021) use a lagged number of store entries (1-week lag) as an IV when analyzing the effect of offline store entries on online purchases. In studying staffing decisions in retail stores, Chuang et al. (2016) use lagged labor (1- and 4-day lags) to address concerns about endogeneity (mostly reverse causality) between labor and sales. Roth et al. (2019) also use lagged independent variables (bed utilization and percent of physicians employed) to investigate how these factors affect a hospital's probability of achieving high performance on clinical quality, the patient experience, and technical efficiency. Besides lagged independent variables, authors frequently use industry averages (or other central tendency measures) as instruments. Astvansh and Jindal (2021) utilize the unanticipated change in the median value of provided trade credit for other firms operating in the focal firm's industry as an instrument, while Golara et al. (2021) employ four IVs, all based on industry averages: sales and after-sales ratings from similar stores within and outside the car dealership state, to prevent a spurious relationship between dealer ratings and the market share.

Going beyond lagged variables and industry averages, Li et al. (2021) employ government social expenditure as an instrument to alleviate concerns about the simultaneity bias between corporate social responsibility and a firm's idiosyncratic risk. McKie et al. (2018) use the sum of observations' characteristics (i.e., total active, feedback score, net feedback score, and return policy) as IVs to alleviate concerns about an endogenous relationship between price and the nested market share.[16] Finally, Muthulingam et al. (2021) investigate the effect of water scarcity on manufacturing's toxic release levels, using the average number of modifications in their manufacturing processes for a given chemical to instrument the modification variables for each chemical.

3.3 | Regression Discontinuity (RD) design

RD is another LATE estimator, which maintains the assumption of selection on observables for a sharp RD (SRD) design but relaxes that assumption for a fuzzy RD design.[17] The RD approach leverages settings where units are eligible for some treatment if some continuous value of a running variable x crosses a threshold c (Angrist & Lavy, 1999; Hahn et al., 2001; Shadish et al., 2002). The units just below the cutoff (that do not receive the treatment) constitute a control group for those just above it (that receive the treatment). For instance, Leung et al. (2020) examine the impact of the labor unionization of customer firms on supplier performance, exploiting RD in close union elections with the share of votes as a running variable. Firms with 49% of the votes for unionization will not be unionized, while those with more than 50% will be. Arguably, if there is no reason to believe that the firms in which the unions barely lost in elections are significantly different from those who barely won, which can make the former a good control group for the latter.

The treatment effect in SRD is estimated as follows:[18]

$$\tau_{SRD} = \lim_{x \downarrow c} E\left(Y(1)\,|X = c\right) - \lim_{x \uparrow c} E\left(Y(0)\,|X = c\right). \quad (13)$$

An advantage of RD is that its identification assumptions are easier to present and evaluate graphically. RD relies on the *continuity assumption* that all potentially relevant variables except for the treatment and outcome are continuous at the cutoff. That is, $E(Y(1)|X, Z)$ and $E(Y(0)|X, Z)$ are continuous in x near the cutoff point $X = c$ (no manipulation assumption). The relevant variables should include the running variable and all the covariates that are considered correlated, with both the outcome and the treatment. A compelling graphical presentation of RD would include plotting the treatment, outcome variable, and important covariates against the running variable and showing a significant jump in the value around the cutoffs for the first two.

However, RD has limitations if units can self-select into treatment based on unobserved factors by manipulating the running variable (i.e., efficiently mobilizing workers to not unionize with a small margin). To address the manipulation issue, the researcher can check whether a plot of the density distribution of the running variable shows a bunching of observations just above the cutoff.[19] Another limitation of the RD approach is that it relies on arbitrary cutoffs in determining the treatment status, which might not be available in most research settings. Also, as RD compares units just above and below the cutoff, it estimates the LATE around the cutoff. As we move further from the threshold, the units will not be comparable. Finally, limiting attention to units just around the cutoff requires a relatively large sample because the estimation is based mainly on the units around the cutoff only.

In recent years (2016–2021), RD has been one of the least employed methods in *POM* articles. It has been used to study the effects of the labor unionization of firms on the operating performance of their suppliers (e.g., Leung et al., 2020). RD design has also been employed to assess the effect of stricter regulations on companies' non-compliance in the auto industry (e.g., Hu et al., 2021) and test the effectiveness of sanctions on the volume of spam produced by companies (e.g., Tang & Whinston, 2020). Beyond *POM*, RD has also been employed to assess the impact of disclosing inventory scarcity on online sales (Park et al., 2020). Park et al. (2020) first show the negative effect of scarcity messages on daily sales using naïve regression but highlight endogeneity concerns when employing this approach. Park et al. (2020) then use the RD design and show that the naïve regression underestimates the negative effect of scarcity messages.

3.4 | Fixed Effects, Differences-in-differences, and Synthetic Control Approaches

So far, we have discussed various empirical strategies that apply to cross-sectional data, in which the fundamental problems of causal inference arise because we cannot observe the same units in both the treatment and control states. However, panel data at least enable us to observe the same unit in the treatment and control states "at different points in time" (Morgan & Winship, 2007). Next, we discuss approaches that are mostly limited to panel data (e.g., FEs, DID, and SC) to estimate counterfactual outcomes based on assumptions regarding the evolution of the outcomes for the control units. These approaches, similar to IV and fuzzy RD, relax the assumption of selection on observables.

3.4.1 | Fixed effects (FE)

A typical specification for panel data takes the following form where i indexes cross-sectional units (e.g., a firm), t indexes time (e.g., a year), and α_i and γ_t are unit (e.g., firm) and time FEs, respectively. X_{it} are other time-varying covariates, and Z_{it} is a time-varying treatment.[20]

$$y_{it} = \beta Z_{it} + \alpha_i + \gamma_t + \delta X_{it} + \varepsilon_{it}. \quad (14)$$

The identification of β depends on the following assumption (Angrist & Pischke, 2009):

$$\mathrm{E}\left(y_{0it}|A_i,\ X_{it},\ \mathrm{t}, Z_{it}\right) = \mathrm{E}(y_{0it}|A_i,\ X_{it},\ \mathrm{t}), \quad (15)$$

where A_i are unobserved factors that are invariant over time (e.g., ability in returns on education in labor economics). This assumption is essentially a slightly relaxed version of the CIA, which states that the treatment Z_{it} is as good as random assignment conditional on specific types of unobserved factors A_i that do not vary over time and other observed covariates X_{it}. The unit FEs capture unobserved time-invariant sources of heterogeneity across units. The year FEs capture macroeconomic shocks common to all units for every period. When the regression includes firm and year FEs (as in two-way FEs [2FEs] models), only the omitted variables that vary by both firm and year would threaten identification.

More specifically, FEs models involve assumptions such as (i) the constant effects of unobserved time-invariant confounders over time, (ii) the absence of unobserved time-varying confounders, (iii) the past outcomes do not affect the explanatory variables (no reverse causality), and (iv) the past values of treatments do not affect the current outcomes (no lagged treatment effects).[21] Depending on the research questions and available data, satisfying these assumptions can be challenging (Angrist & Pischke, 2009; Cunningham, 2021; Firebaugh et al., 2013; Halaby, 2004; Hill et al., 2020; Morgan & Winship, 2007).

The FEs approach is the most frequently used identification strategy employed in recently published *POM* papers. These *POM* papers use unit FEs to control for time-invariant unit-level unobserved factors, such as industry, location, and long-term production capabilities (e.g., Ayer et al., 2019; Bradley et al., 2018) or state and year FE (e.g., Drake & York, 2021) to account for yearly industry-specific or state-specific macroeconomic shocks.

3.4.2 | Differences-in-differences (DID)

The simplest form of the DID design is a 2×2 design with a treated group T and an untreated group C, in which we observe outcomes both before and after the treatment for both groups. It can be written as the *difference* between the two *time differences* in the outcomes (post–pre) among the treated and control groups.

$$\hat{\beta}_{DD} = (E[Y_T|Post] - E[Y_T|Pre]) - \left(E[Y_C|Post] - E[Y_C|Pre]\right). \quad (16)$$

When the treatment variable is time-varying and binary (e.g., adoption of ERP), one can apply the DID approach with panel data using the following specification to estimate $\hat{\beta}_{DD}$:

$$y_{it} = \beta_{\mathrm{DD}} Treat_i * Post_t + Treat_i + Post_t + \varepsilon_{it}. \quad (17)$$

In this specification, $Treat_i$ is a dummy variable equal to one if firm i is treated (e.g., adopts an ERP system), $Post_t$ is a dummy variable equal to one if year t is after the treatment, and β_{DD} is the causal effect of the binary treatment.

Identification in a DID specification critically hinges upon the parallel trends assumption. As an example, the identification assumption for DID regarding ERP impact is that in the absence of the treatment (i.e., ERP adoption), the *changes* in the outcomes among the treated firms will be the same as the *changes* in the outcomes among the control firms, although these two groups might have different starting levels in the outcomes. More formally, we can rewrite $\hat{\beta}_{DD}$ following the potential outcomes framework as the sum of ATT and a term that captures the deviation from the parallel trends assumption (Cunningham, 2021):

$$\hat{\beta}_{DD} = \left(E[Y_T^1|Post] - E[Y_T^0|Post]\right) + \left(E[Y_T^0|Post] - E[Y_T^0|Pre]\right) - \left(E[Y_C^0|Post] - E[Y_C^0|Pre]\right). \quad (18)$$

An effective way to convince readers about this assumption is to visually assess whether the pre-treatment trends between the treated and control groups evolve in parallel. Formally, we could test for the parallel in the pre-trends in an event study analysis/relative time model, where the mean difference in the outcomes between the treated and control units is estimated period-by-period, with several periods before and after the treatment (Marcus & Sant'Anna, 2021).

Unlike the simple 2×2 DID design, which allows only one treatment time, a more general version of DID can accommodate multiple treatment cohorts in the following 2FEs (TWFE or 2FE) model:

$$y_{it} = \beta_{\mathrm{DD}} Treat_{it} + \alpha_i + \delta_t + \varepsilon_{it}. \quad (19)$$

In this specification, $Treat_{it}$ is a dummy variable that varies at the firm-year level and assumes the value of one if firm i has been treated. This specification also allows firms to be treated in different periods.[22] A special case of regression DID (Angrist & Pischke, 2009) is "staggered DID," which assumes that once a treatment such as ERP is adopted, it will remain adopted by each adopting firm (e.g., once adopted, $Treat_{it}$ will assume the value of one in all subsequent periods). For example, Wang et al. (2022) study the impact of additional airline routes on cadaveric kidney transplants, utilizing a DID approach with staggered treatment adoption.[23]

The DID approach is frequently used in empirical studies recently published in *POM*, often in conjunction with FEs. For instance, researchers have employed the DID approach to investigate the effect of distance to households on Costco's resilience to competition (e.g., Lim et al., 2021) and buy-online-pickup-in-store effects on future purchase frequencies and amounts (e.g., Song et al., 2020).

3.4.3 | Synthetic Control (SC) methods

The SC method is related to DID in that it allows adjustment for the pre-treatment characteristics, especially when there are one treated unit and many more control units (Abadie & Gardeazabal, 2003; Abadie et al., 2010).[24] The SC method consists of a combination of the control units to constitute a more appropriate comparison than any single unaffected unit alone. This method weights the control units and uses the trend of the "weighted" control units as a plausible counterfactual unit into which the treated unit would evolve had the treatment never been applied. Using the single-treated unit ($j = 1$) with multiple control units ($j = 2, 3, \ldots, J$) as an example, the synthetic control estimator is:

$$\hat{\tau}_{1t} = Y_{1t} - \sum_{j=2}^{J} w_j^* Y_{jt}, \tag{20}$$

where W^* is estimated by minimizing the norm $\|X_1 - X_0 W\| = \sqrt{(X_1 - X_0 W)' V (X_1 - X_0 W)}$, $W^* = (w_2^*, \ldots, w_J^*)'$. X_1 and X_0 are vectors of pre-treatment characteristics for the treated and control units, respectively. Equation 20 illustrates how the SC procedure constructs the counterfactual post-treatment outcome of the treated units had the treatment never occurred, using pre-treatment data. The first step is to obtain the weights W^* that make the weighted average of pre-treatment covariates of the control units (X_0) best mimic those of the treated units (X_1), minimizing the aforementioned norm. Since there are usually multiple covariates, their relative importance in the norm is defined by another set of weights V, and W^* depends on V. Abadie et al. (2010) offer a variety of options in choosing V. In practice, many researchers choose V to minimize the mean squared prediction error of the pre-treatment (up to T_0) outcomes (i.e., $\sum_{t=1}^{T_0} (Y_{1t} - \sum_{j=2}^{J} w_j^*(V) Y_{jt})^2$). Then, one can calculate the weighted average of the post-treatment outcome of the control units with the same set of weights (i.e., the second term in the right-hand side of Eqn 20), and use it as the counterfactual outcome of the treated units in the absence of the treatment.

One of the main differences between the DID and SC methods is that DID is usually applied in cases with many treated units, while SC is applied to cases with only a single or a few treated units. Also, while DID critically relies on the "parallel trends" assumption, SC relaxes this assumption. Even if the average trends between the treated and control units diverge, one could reweight the control units to match their pre-trends to those of the treated units (Arkhangelsky et al., 2021). Note that optimal control should approximately match the intervention unit on two dimensions: significant pre-intervention predictors of the outcome and the pre-intervention outcome.

One of the main attractive features of SC is that it produces appealing figures: with a relatively rich pool of control units and relatively long pre-treatment time series, SC could produce a pre-treatment time trend among the "synthetic" control group as close to that of the treated group as possible. This helps present the estimated treatment effect graphically as divergent lines from the treatment period, which helps communicate the identification more intuitively. By comparing the treatment effect sizes against the placebo setups, SC can generate statistics comparable to significance tests. However, as a comparative case study method, it does not allow for traditional significance tests. Instead, it verifies the treatment effect through placebo tests, such as a permutation of the treated unit and treatment time (Abadie et al., 2015). A few requirements are imposed on the data to achieve good performance in SC. First, SC relies on large pre-intervention periods to train. The longer the pre-intervention period, the better (Abadie et al., 2010), and applying SC to a short pre-intervention period might lead to over-interpolation. Second, SC requires balanced panel data to train the weights.

Among the *POM* applications of SC, Adbi et al. (2019) study the effect of the 2009–2010 global H1N1 pandemic as an exogenous demand shock on vaccine manufacturers in India and report that the pandemic shock caused a decrease in the market share of multinational companies and an increase in the market share of domestic companies. They create an SC group based on a weighted average of the pre-pandemic period outcomes from 19 vaccine manufacturers that do not supply influenza vaccines as the counterfactual comparison group to firms that manufacture influenza vaccines. In addition to SC, Adbi et al. (2019) also use CEM and two other alternative control groups to verify the consistency of their results.

3.5 | Summary of identification strategies used in *POM* papers

Table 1 summarizes how alternative approaches for estimating causal effects address the two challenges related to the baseline and differential treatment effect biases in assessing causality and their limitations due to either data characteristics or inherent untestable assumptions. Sometimes, using a particular approach also limits the types of causal questions that researchers can answer. An astute reader will note that researchers often estimate far narrower effects or effects that are more difficult to interpret for any particular group of units in their sample. Such limitations are not particular to quasi-experimental studies; even randomized controlled trials, often held as exemplars for assessing causality, raise serious concerns (Deaton & Cartwright, 2018), even when we ignore non-compliance and attrition, for instance.

Fortunately, OM researchers sometimes employ more than one identification strategy in their studies. Among the recently published studies in *POM* (2016–2021) that have used identification strategies, 49% have employed more than one strategy.[25]

TABLE 1 How different identification strategies solve key challenges in assessing causality[a]

Approach	How baseline bias is addressed	How differential treatment effect bias is addressed	What is estimated	Limitations
1. Regression-based	Assumes strong ignorability (i.e., selection on observables only)	Ruled out by constant-coefficient models	ATE	Assumption of correct specification of the relationship between Y and the controls. No distinction between the treatment and covariates
2. Matching and weighting	Assumes strong ignorability (i.e., selection on observables only)	Possible to assess differential treatment effects across strata	ATT, ATU, or ATE (depends on the assumptions)	It is possible to conduct sensitivity analyses (e.g., Rosenbaum's gamma) for assessing selection bias and for the violation of the strong ignorability assumption
3. Instrumental variable (IV)	Relaxes the assumption of selection on observables. Exploits randomization induced by the IV	Ignores treatment heterogeneity by estimation for only a subgroup	LATE (Only for compliers or defiers, but not both)	Difficult to find strong and relevant IVs. Exclusion restriction (IV affects the outcome only via treatment) is not testable. Yields large standard errors if the sample sizes are small
4. Regression discontinuity (RD)	Sharp RD uses the assumption of selection on observables, but fuzzy RD relaxes that. Exploits the randomization induced by the cutoff score on the running variable	Ignores treatment heterogeneity by estimation for only a subgroup	LATE	Running variable perfectly (or fuzzily) determines the treatment assignment. Assumes no discontinuity in other factors that could also affect the outcome other than the treatment
5. Differences-in-differences (DID)	Exploits within-unit variation over time, assuming that all unobservables are time-invariant	Ignores treatment heterogeneity between ATT and ATU	ATT	Assumes parallel trends: Had there been no treatment, the trend in the outcomes would be parallel between the treated and the control. Mostly useful for sharp binary interventions. Requires longitudinal data on both the treated and control units
6. Fixed effects (FEs)	Exploits within-unit variation over time, assuming that all unobservables are time-invariant. Uses changes over time in the control group as a counterfactual for the changes in the treated group	Ignores treatment heterogeneity (assumes that the heterogeneity of the unit-specific causal effect across the population is random)	ATT	Needs strong assumptions or long time series for modeling the counterfactual trajectory. Assumes that past outcomes do not influence the treatment and no lagged treatment effects

[a]Figure B in Appendix B of the Supporting Information contrasts different approaches

4 | DISCUSSION

Our goal in this article was to review some dominant identification strategies increasingly used in OM research for causal inference with observational data. Counterfactual, or potential outcomes, approaches help focus on the "design" aspects of studies, with careful attention to "choices" such as (i) a better way of testing broad and competing theories using specific, focused, and narrower research questions or hypotheses; (ii) constructing treated and control groups with abrupt, short-lived treatments; (iii) considering multiple treatment assignment mechanisms; (iv) nondose nonresponse analyses; (v) the use of natural blocks (e.g., twins, siblings, schools, neighborhoods); and (vi) minimizing the need for stability analyses that assess the impact of innocuous changes in analytical decisions on conclusions (Rosenbaum, 1999, 2002). This emphasis on "design" or "choices" can be contrasted with relying on statistical controls or many functional form assumptions in the traditional econometrics literature. We discussed how different approaches provide their vantage point for assessing the causal effect that critically depends on the assumptions of that approach. We find that no one approach is unconditionally superior to another in all circumstances, reminding us of the value of triangulation when using multiple approaches across studies.

We next discuss some issues that often arise in assessing causality based on our own experience of using these methods as authors, reviewers, or editors, and the types of conversations that unfold during the review processes to provide some opportunities for clarification and reflection by all participants in academic discourse.

First, although assessing causality is an important part of academic research, it is just one aspect of an important contribution for some papers, and other issues such as novelty and the framing of new research questions, developing new theories, mechanisms, or methods to understand operational and management issues are no less important. Note also that causal assessment in quasi-experimental studies (or even in experimental studies) have degrees of plausibility, and by itself, no one study ever proves causation, no matter how many robustness checks one conducts or how clever one's research design is. There seems to be agreement among the proponents and detractors of the instrumental, natural, and randomized experimental-based approaches, sometimes labeled as "atheoretic" (Deaton, 2010; Imbens, 2010; Keane, 2010), that such methods are appropriate only for certain questions, and the credibility of these methods is "conditional on the question." However, it would be a pity if researchers "avoid questions where randomization is difficult, or even conceptually impossible, and natural experiments are not available" (Imbens, 2010, p. 401). Even the best experimental or quasi-experimental studies only inform us of a particular effect (whether that effect is present or absent and its magnitude); it rarely informs us of the underlying mechanisms without carrying out further experiments or further studies in other settings. In other words, the type of causation revealed by quasi-experimental studies is "molar" causation and not the "molecular" causation that provides a deeper understanding of why effects occur in the first place (Shadish et al., 2002).

Second, even if one strongly subscribes to the potential outcomes view of causality, within that, there are multiple ways of estimating "causal effects," such as those based on (i) conditioning on variables (e.g., regression and matching/weighting), (ii) exogenous variation in an IV to isolate covariation in the causal and outcome variables, (iii) the use of isolated and exhaustive mechanisms[26] that relate the causal variable to the outcomes, and (iv) approaches based on longitudinal data (Morgan & Winship, 2007). All identification strategies rely on their assumptions, and sometimes there is an unrealistic expectation that one particular method (e.g., FEs or DID, depending on a reviewer's preference) always yields better or more correct answers than some other method that authors may prefer to use in a study. Although reasonable disagreements on preferences may remain, reviewers' arguments for their preferences must be justified using the same standards to which authors are subjected for a balanced perspective on the advantages and limitations of particular approaches.

Third, because most of the identification approaches using potential outcomes focus only on binary treatments, it does not mean that researchers should ignore research opportunities when they suspect significant interactions, curvilinear or multi-level effects, or those in which the interventions are best viewed as configurations. Therefore, OM scholars must also remain open to other ways of assessing causality, such as that based on a configurational logic with necessary and sufficiency conditions (Arellano et al., 2021; Mahoney et al., 2013; Park & Mithas, 2020), which admit notions of conjunctural, asymmetric, and equifinal causation. Also, they should consider correlational/covariance-based approaches that illuminate "causal" pathways that are increasingly being reinterpreted as a valid mechanism-based approach for causal inference in a potential outcome approach under certain conditions (Barringer et al., 2013; Bollen & Pearl, 2013; Morgan & Winship, 2007; Shah & Goldstein, 2006).[27] OM scholars should consider the use of bounds (Manski & Nagin, 1998), sensitivity analyses (Rosenbaum & Rubin, 1983a), and other approaches (Frank, 2000) that express the uncertainty of estimates in terms of observables or correlations known to scientists in that domain versus overreliance on assumptions about disturbances, functional forms, or exclusion restrictions.[28]

We see great value in assessing causality via careful and detailed longitudinal case studies or by using "shoe-leather," as John Snow did to establish causation for his studies of the causes of cholera more than 150 years ago, instead of "statistics" alone (Freedman, 1991). No amount or type of data (e.g., Big data, thick data, dark data) or statistics and analytics are a substitute for selecting significant problems, posing important and substantive questions, careful and deliberate data collection, careful thinking of the assumptions underlying methods and their applicability in particular settings based on substantive domain knowledge that comes about by studying the same phenomenon over time across settings using different logics (e.g., mediation vs. moderation) and mechanisms (Goldthorpe, 2001), and assessing generalizability via complex pattern matching across several studies (Shadish et al., 2002). Indeed, the choice of a particular approach should depend on the research questions, stage of research (e.g., exploratory, descriptive, explanatory, predictive), and type of data to which one has access rather than choosing trivial problems just because they use a new "quasi-experimental" method or what some call a "low-grade quasi-experiment" (Shadish et al., 2002).[29]

To conclude, this article provides an overview of commonly used identification approaches for causal inference used in most *POM* papers. Increasingly, *POM* papers are using relatively newer potential outcomes-based approaches for assessing causality that coincides with similar trends in other business disciplines (The Economist, 2016). Science in general and business research are no exceptions; they thrive on the plurality of methods of inquiry (Agrawal et al., 2020) but also admonish scientists to be skeptical of all that is new or claimed to be so. Francis Bacon, a 17th-century philosopher of science, warned: "We must steer a middle course between our appetites for the old and our love of the new. Both can deceive us." (quoted in Wandersee, 1990, p. 95). Given the domain variety and constantly expanding frontiers of research in OM in the face of emerging technologies and methods (Mithas, Chen, et al., 2022; Tang & Veelenturf, 2019), we remain optimistic that OM researchers will also embrace other notions of causality and ways of gaining understanding via careful descriptive research. Such research may eventually pave the way for associational studies, studies with causal mechanisms, and studies that exploit exogenous shocks and natural experiments, finally leading to more complex structural models for a better understanding of the interactions among men, technologies, markets, and social institutions for desirable outcomes.

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ENDNOTES

[1] Appendix A in the Supporting Information presents a bibliometric analysis of UTD-24 journals and a more detailed analysis of empirical papers published in *POM* from 2016 to 2021. We identified 122 papers that were causal, out of 197 causal/associational, 411 empirical papers, and a total of 1017 papers published in *POM* during 2016–2021. In other words, the proportion of papers that used one of the identification approaches such as matching, regression discontinuity, difference-in-difference, FEs, or IVs over all papers published in *POM* from 2016 to 2021 was about 12.5%.

[2] Figure A4 in Supporting Information shows a Pareto analysis in which FEs, IVs, and matching together accounted for 85% of the identification strategies present in *POM* papers during the 2016–2021 period. Table A2 in the Supporting Information shows combinations of identification strategies in *POM* papers from 2016 to 2021. During these 6 years, FEs has been the most popular strategy, and its use has grown substantively from 2020 to 2021. The use of IVs, matching, and difference-in difference also shows an upward trend.

[3] These effects include ATE, ATT, ATU, MTE, and LATE.

[4] We prefer terms such as *baseline bias* and differential treatment effect bias (aka treatment effect heterogeneity) here (Winship & Morgan, 1999; Xie, 2011) because they are clearer conceptually to other terms such as endogeneity bias or selection bias, which over time have been used to describe a variety of problems in causal inference. We have ourselves been guilty of using terms such as endogeneity or selection bias in our writing (e.g., Mithas & Krishnan, 2009; Mithas et al., 2014) when we should have used more descriptive terms such as the ones used here.

[5] A related concept is reverse causality, which happens when Y causes X.

[6] Researchers frequently use contemporaneous covariates in place of pre-treatment covariates due to a lack of pre-treatment data and try to rule out concerns due to reverse causality by invoking assumptions or theories.

[7] The identification of causal effects in the potential outcomes approach also invokes the Stable Unit Treatment Value Assumption (Angrist et al., 1996), that is, the potential outcomes of a particular unit is uncorrelated with the treatment status of other units.

[8] Note that a regression-based approach is motivated by an "attempt to condition on all other relevant direct causes of the outcome," while matching is motivated by "an attempt to balance the variables that predict treatment assignment/selection" (Morgan & Winship, 2007, p. 50). The regression formulation differs from the matching estimators in that with regressions, we make additional modeling assumptions such that we know the true model for the mean of Y conditional on Xs (e.g., the linear relationship in Equation 6). One problem with this approach is that we may misspecify the model for the mean of Y given the Xs, thereby producing biased estimates for the causal effect. This problem would persist even if the CIA were satisfied. Therefore, not only is the choice of covariates crucial (to satisfy the CIA) but also the choice or correctness of the presumed model for the conditional mean of Y given Z and X. The matching and stratification-based estimators do not require models for the conditional mean of Y.

[9] Essentially, this assumption says that selection (and thus, confounding bias) takes place only on the observables X; that is, given information on X alone, we can obtain causal effect estimates as we would in a randomized experiment. An implication of the strong ignorability or CIA is that matching does not solve the problem arising from potential unobserved factors that correlate with both the treatment and outcomes. There are a few approaches that allow one to conduct sensitivity analyses to assess the extent to which the matching estimators are sensitive to the assumption of selection on unobservables (Rosenbaum, 2002) or to bound the parameter estimates with weaker assumptions about selection into treatment and the response to the treatment (Manski & Nagin, 1998).

[10] Some studies suggest that if we keep pruning data beyond the common support of the propensity score, the imbalance in the covariates will decrease in the beginning but might increase at some point (King & Nielsen, 2019).

[11] As an example, if we have four continuous covariates, we could divide each one into three groups to end up with 3*3*3*3= 81 strata. After creating these strata, the researcher could choose to: (1) match each treated firm to all control firms within the same strata, and exclude firms assigned to strata with no treated firms; or (2) assign weights to the control firms within matched strata accounting for the number of observations within the strata or other criteria, with treated firms always assigned a weight of one and the units in the strata with either no treated firms or no control firms with the weight of zero. Note that CEM drops treated units not matched to any control units within a strata, and researchers can adjust the number of strata to avoid dropping too many treated units.

[12] Other recently published papers in *POM* using PSM also present such distribution balance checks (Xue et al., 2019; Zhang & Yao, 2020).

[13] Arguably, the intrinsic value of the organizer is not observed in this example. For causal identification, it is assumed that the intrinsic value is correlated with observables such as the campaign target, and the remaining sources of unobservables would not be correlated with the treatment. Further examples of CEM can be found in other papers recently published in *POM* (e.g., Avgerinos & Gokpinar, 2018; Karimi et al., 2021).

[14] Recent work by Saldanha et al. uses the hardware investments of firms as one of the instruments for information technology (IT) investment (Saldanha et al., 2020), with the logic that hardware investments are likely to be highly correlated with IT investment. However, by themselves and in the absence of software investments, hardware investments are unlikely to directly influence firm performance except through IT investments (Rai et al., 1997). Researchers often use the lagged value of the treatment or industry averages as instruments; the validity of these IV choices is context dependent, and oftentimes a matter of debate. For example, an arguably lagged treatment may not only affect the current treatment but also the outcome Y if the treatment effects take time to materialize, thereby violating the exclusion restriction. The use of industry averages or leave-out means also depends on specific exclusion restriction assumptions: there should be no spillover effects (the treatment status of other firms in the same industry affects the outcome of the focal firm) nor interdependence effects (the outcomes of other firms affect that of the focal firm, likely through market competition). Betz et al. (2018) discuss the usage and limitations of such group average instruments (or spatial instruments).

[15] The LATE framework divides a population into four mutually exclusive groups: (1) Compliers are the units whose treatment status is affected by the instrument in the expected direction, (2) defiers are those whose treatment status is affected by the instrument in the wrong/opposite direction, (3) always-takers, who will always be treated irrespective of the instrument, and (4) never-takers, who will always remain untreated irrespective of the instrument. Defiers are often assumed away by the monotonicity condition, as they are rare and correspond to somewhat irrational behavior. Always-takers and never-takers play no role in IV estimation. It is often difficult to tell if a specific treated unit is a complier or an always-taker because these groups are latent and unobserved. In other words, LATE is not ATT, barring a few exceptions, because ATT is a weighted average of effects on always-takers and compliers (Angrist & Pischke, 2009).

[16] A common identification problem in demand estimation is that prices are endogenously determined by firms because unobserved market factors may push demand and firms' price responses simultaneously. Classic instruments for prices are cost shifters such as exchange rate shocks, input prices, or the characteristics of competing products—sometimes called "Berry–Levinsohn–Pakes (BLP) instruments" (Berry et al., 1995).

[17] Angrist and Pischke (2009) note: "The sharp design can be seen as a selection-on-observables story. The fuzzy design leads to an IVs type of setup."

[18] Here, we mainly focus on sharp regression discontinuity (RD) because the only OM paper we survey uses sharp RD. Another type of RD is fuzzy RD, where $\lim_{x\downarrow c} p(T = 1|X = x) \neq \lim_{x\uparrow c} p(T = 1|X = x)$, such as in W. Zhao et al. (2021). That is, incentives to participate in a treatment may change discontinuously at a threshold c, but the incentives are not enough to guarantee that all units will be treated when x crosses the threshold c.

[19] A useful tool is the Rddensity package developed by Cattaneo et al. (2018).

[20] In many cases, Z_{it} might be binary variables (e.g., ERP adoption) and could be analyzed with a DID framework (often in conjunction with FEs). In this section, we discuss in general the FEs approach with either binary or continuous treatments.

[21] Assumption (iv) could be relaxed by adding lagged treatments ($X_{i,t-1}$, $X_{i,t-2}$, etc.) in the same regression. However, the exact number of lagged treatments are often unknown in the absence of strong theories. An exception is when the treatment variable switches one time in one direction (e.g., a staggered DID treatment, where the treatment variable switches once from untreated to treated and remains treated afterward; e.g., Imai & Kim, 2019).

[22] The traditional TWFE estimator might be biased in the case of strong dynamic treatment effects. There are new econometrics tools available to deal with this issue. Goodman-Bacon (2021) offers a useful decomposition of the two-way FEs estimator β_{DD} as a weighted average of all potential 2×2 DID estimates across different comparion groups (treated vs. untreated, early vs. later treated, later vs. early treated, etc.) and suggests that β_{DD} could be biased in some settings. Callaway and Sant'Anna (2021) and Sun and Abraham (2021) propose alternative estimators that adjust for the biases mentioned above. Recent work provides new insights on the assumptions of unit FEs (e.g., Imai & Kim, 2019) and two-way FEs models (e.g., Imai & Kim, 2021) when one focuses attention on binary treatments. These new techniques can allow researchers to choose appropriate methods, depending on the assumptions about the relative importance of causal dynamics and time-invariant unobserved confounders. We offer a more detailed discussion of the issues in Appendix B.

[23] Some studies also use triple difference (or difference-in-difference-in-differences [DDD]) estimator, which is effectively the difference between two DID estimators (Cunningham, 2021). A classic example is Gruber (1994)'s examination of maternity benefits mandates on labor market outcomes. Beyond the traditional DD setting (mandates passed in treated states), the author compares differential "treatment effects" of the same law between exposed (i.e., women of childbearing age) and unexposed groups. One can view DDD as a falsification exercise, and most of the causal effect estimate in this design comes from changes in the treatment units, instead of changes in the control units.

[24] An extension allows for more than one unit to experience treatment and at possibly different times (Cavallo et al., 2013).

[25] For a more detailed analysis of the combinations, such as the strategies and examples of papers, see Appendix A in the Supporting Information. Figure A4 shows that among the 122 papers that employed any identification strategy, 65 have utilized only one strategy, which implies that roughly half (47%) of the papers have used two or more strategies to assess causality. More specifically, 33 papers have used two, 18 have used three, and 6 have used four strategies.

[26] Mediating variables should completely account for the causal effect of treatment on outcome variable, and the mediating variables should only be influenced by treatment (and not any other factor; Morgan & Winship, 2007). These are fairly stringent assumptions that are rarely met in practice (Zhao et al., 2010).

[27] We see a role here for editors to step in if authors are pushed toward conforming to particular preferences of reviewers rooted either in strong preference for assumptions that they believe or in some cases ignorance or intolerance of other "causal" methods. We also caution against relying solely on reviewers' judgments of papers when reviewers' comments indicate extreme skepticism such as asking authors to treat all of their explanatory variables as endogenous (there is no current method that addresses that issue at least at the time of this writing)!

[28] Very few *POM* studies have so far used these approaches (Khuntia et al., 2021).

[29] Appendix B in the Supporting Information provides a brief description of emerging machine learning-based methods for causal inference. Very few papers have so far used machine learning (ML) methods for causal inference in OM studies, barring some exceptions (Miao et al., 2021).

REFERENCES

Abadie, A., Diamond, A., & Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. *Journal of American Statistical Association*, *105*(490), 493–505.

Abadie, A., Diamond, A., & Hainmueller, J. (2015). Comparative politics and the synthetic control method. *American Journal of Political Science*, *59*(2), 495–510.

Abadie, A., & Gardeazabal, J. (2003). The economic costs of conflict: A case study of the Basque country. *American Economic Review*, *93*(1), 112–132.

Achen, C. H. (2002). Toward a new political methodology: Microfoundations and ART. *Annual Review of Political Science*, *5*, 423–50.

Adbi, A., Chatterjee, C., Drev, M., & Mishra, A. (2019). When the big one came: A natural experiment on demand shock and market structure in India's influenza vaccine markets. *Production and Operations Management*, *28*(4), 810–832.

Agrawal, V., Khanna, P., & Singhal, K. (2020). The role of research in business schools and the synergy between its four subdomains. *Academy of Management Review*, *45*(4), 896–912.

Angrist, J. D., Imbens, G. W., & Rubin, D. B. (1996). Identification of causal effects using instrumental variables. *Journal of American Statistical Association*, *91*(434), 444–455.

Angrist, J. D., & Lavy, V. (1999). Using Maimonides' rule to estimate the effect of class size on scholastic achivement. *Quarterly Journal of Economics*, *114*, 533–75.

Angrist, J. D., & Pischke, J.-S. (2009). *Mostly harmless econometrics: An empiricist's companion*. Princeton University Press.

Arellano, M. C, Meuer, J., & Netland, T. H. (2021). Commitment follows beliefs: A configurational perspective on operations managers' commitment to practice adoption. *Journal of Operations Management*, *67*(4), 450–475.

Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., & Wager, S. (2021). Synthetic difference-in-differences. *American Economic Review*, *111*(12), 4088–4118.

Astvansh, V., & Jindal, N. (2021). Differential effects of received trade credit and provided trade credit on firm value. *Production and Operations Management*, *2*, 781–798.

Avgerinos, E., & Gokpinar, B. (2018). Task variety in professional service work: When it helps and when it hurts. *Production and Operations Management*, *27*(7), 1368–1389.

Ayer, T., Ayvaci, M. U. S., Karaca, Z., & Vlachy, J. (2019). The impact of health information exchanges on emergency department length of stay. *Production and Operations Management*, *28*(3), 740–758.

Ba, S., He, S., & Lee, S. Y. (2022). Mobile app adoption and its differential impact on consumer shopping behavior. *Production and Operations Management*, *31*(2), 764–780.

Barringer, S. N., Eliason, S. R., & Leahey, E. (2013). A history of causal analysis in the social sciences. In S. L. Morgan (Ed.), *Handbook of causal analysis for social research* (pp. 9–16). Springer.

Berry, S., Levinsohn, J., & Pakes, A. (1995). Automobile prices in market equilibrium. *Econometrica*, *63*(4), 841–890.

Betz, T., Cook, S. J., & Hollenbach, F. M. (2018). On the use and abuse of spatial instruments. *Political Analysis*, *26*(4), 474–479.

Bollen, K. A., & Pearl, J. (2013). Eight myths about causality and structural equation models. In S. L. Morgan (Ed.), *Handbook of causal analysis for social research* (pp. 301–328). Springer.

Bound, J., Jaeger, D. A., & Baker, R. M. (1995). Problems with instrumental variables estimation when the correlation between the instruments and endogenous explanatory variables is weak. *Journal of American Statistical Association*, *90*(430), 443–450.

Bradley, R. V., Esper, T. L., In, J., Lee, K. B., Bichescu, B. C., & Byrd, T. A. (2018). The joint use of RFID and EDI: Implications for hospital performance. *Production and Operations Management*, *27*(11), 2071–2090.

Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods and an application on the minimum wage and employment. *Journal of Econometrics*, *225*(2), 200–230.

Cascio, E. U., & Lewis, E. G. (2006). Schooling and the Armed Forces Qualifying Test: Evidence from school-entry laws. *Journal of Human Resources*, *41*(2), 294–318.

Cattaneo, M. D., Jansson, M., & Ma, X. W. (2018). Manipulation testing based on density discontinuity. *Stata Journal*, *18*(1), 234–261.

Cavallo, E., Galiani, S., Noy, I., & Pantano, J. (2013). Catastrophic natural disasters and economic growth. *The Review of Economics and Statistics*, *95*(5), 1549–1561.

Chang, H., Liu, S., & Mashruwala, R. (2022). Customer bargaining power, strategic fit, and supplier performance. *Production and Operations Management*, *31*(4), 1492–1509.

Chuang, H. H. C., Oliva, R., & Perdikaki, O. (2016). Traffic-based labor planning in retail stores. *Production and Operations Management*, *25*(1), 96–113.

Crown, W. H., Henk, H. J., & Vanness, D. J. (2011). Some cautions on the use of instrumental variables estimators in outcomes research: How bias in instrumental variables estimators is affected by instrument strength, instrument contamination, and sample size. *Value in Health*, *14*(8), 1078–1084.

Cunningham, S. (2021). *Causal inference: The mixtape*. Yale University Press.

Deaton, A. (2010). Instruments, randomization, and learning about development. *Journal of Economic Literature*, *48*(2), 424–455.

Deaton, A., & Cartwright, N. (2018). Understanding and misunderstanding randomized controlled trials. *Social Science & Medicine*, *210*, 2–21.

Dehejia, R. H., & Wahba, S. (1999). Causal effects in nonexperimental studies: Reevaluating the evaluation of training programs. *Journal of American Statistical Association*, *94*(448), 1053–1062.

Drake, D. F., & York, J. G. (2021). Kicking ash: Who (or what) is winning the "war on coal"? *Production and Operations Management*, *30*(7), 2162–2187.

Firebaugh, G., Warner, C., & Massoglia, M. (2013). Fixed effects, random effects, and hybrid models for causal analysis. In S. L. Morgan (Ed.), *Handbook of causal analysis for social research* (pp. 113–132). Springer.

Frank, K. A. (2000). Impact of a confounding variable on a regression coefficient. *Sociological Methods & Research*, *29*(2), 147–194.

Freedman, D. A. (1991). Statistical models and shoe leather. *Sociological Methodology*, *21*, 291–313.

Golara, S., Dooley, K. J., & Mousavi, N. (2021). Are dealers still relevant? How dealer service quality impacts manufacturer success. *Production and Operations Management*, *30*(10), 3560–3078.

Goldthorpe, J. H. (2001). Causation, statistics and sociology. *European Sociological Review*, *17*(1), 1–20.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, *225*(2), 254–277.

Gow, I. D., Larcker, D. F., & Reiss, P. C. (2016). Causal inference in accounting research. *Journal of Accounting Research*, *54*(2), 477–523.

Greene, W. H. (2000). *Econometric analysis* (4th ed.). Prentice Hall.

Gruber, J. (1994). The incidence of mandated maternity benefits. *American Economic Review*, *84*(3), 622–641.

Gupta, S., Verma, R., & Victorino, L. (2006). Empirical research published in *Production and Operations Management* (1992–2005): Trends and future research directions. *Production and Operations Management*, *15*(3), 432–448.

Hahn, J., van der Klauww, W., & Todd, P. (2001). Identification of treatment effects by regression discontinuity design. *Econometrica*, *69*(1), 201–209.

Hainmueller, J. (2012). Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. *Political Analysis*, *20*(1), 25–46.

Halaby, C. N. (2004). Panel models in sociological research: Theory into practice. *Annual Review of Sociology*, *30*, 507–544.

Heckman, J. J. (1997). Instrumental variables: A study of implicit behavioral assumptions used in making program evaluations. *Journal of Human Resources*, *32*(3), 441–462.

Hill, T. D., Davis, A. P., Micah Roos, J., & French, M. T. (2020). Limitations of fixed-effects models for panel data. *Sociological Perspectives*, *63*(3), 357–369.

Ho, T. H., Lim, N., Reza, S., & Xia, X. (2017). OM forum—Causal inference models in operations management. *Manufacturing and Service Operations Management*, *19*(4), 509–525.

Hu, K., Chopra, S., & Chen, Y. (2021). The effect of tightening standards on automakers' non-compliance. *Production and Operations Management*, *30*(9), 3094–3115.

Imai, K., & Kim, I. S. (2021). On the use of two-way fixed effects regression models for causal inference with panel data. *Political Analysis*, *29*, 405–415.

Imai, K., & Kim, I. S. (2019). When should we use linear unit fixed effects regression models for causal inference with longitudinal data. *American Journal of Political Science*, *63*(2), 467–490.

Imbens, G. W. (2010). Better LATE than nothing: Some comments on Deaton (2009) and Heckman and Urzua (2009). *Journal of Economic Literature*, *48*(2), 399–423.

Karimi, A., Mishra, A., Natarajan, K. V., & Sinha, K. K. (2021). Managing commodity stock-outs in public health supply chains in developing countries: An Empirical analysis. *Production and Operations Management*, *30*(9), 3116–3142.

Keane, M. P. (2010). Structural vs. atheoretic approaches to econometrics. *Journal of Econometrics*, *156*, 3–20.

Ketokivi, M., & McIntosh, C. N. (2017). Addressing the endogeneity dilemma in operations management research: Theoretical, empirical, and pragmatic considerations. *Journal of Operations Management*, *52*, 1–14.

Khuntia, J., Mayya, R., Mithas, S., & Agarwal, R. (2021). Managing cellphone services for customer satisfaction: Evidence from the base-of-the-pyramid markets. *Production and Operations Management*, *30*(2), 438–450.

King, G., & Nielsen, R. (2019). Why propensity scores should not be used for matching. *Political Analysis*, *27*(4), 435–454.

Leung, W. S., Li, J., & Sun, J. (2020). Labor unionization and supply-chain partners' performance. *Production and Operations Management*, *29*(5), 1325–1353.

Li, G., Li, N., & Sethi, S. P. (2021). Does CSR reduce idiosyncratic risk? Roles of operational efficiency and AI innovation. *Production and Operations Management*, *30*(7), 2027–2045.

Lim, S. F. W. T., Rabinovich, E., Park, S., & Hwang, M. (2021). Shopping activity at warehouse club stores and its competitive and network density implications. *Production and Operations Management*, *30*(1), 28–46.

Lu, G., Ding, X. D., Peng, D. X., & Chuang, H. H. -C. (2018). Addressing endogeneity in OM research: A literature review and practical guide. *Journal of Operations Management*, *24*, 53–64.

Mahoney, J., Goertz, G., & Ragin, C. C. (2013). Causal models and counterfactuals. In S. L. Morgan (Ed.), *Handbook of causal analysis for social research* (pp. 75–90). Springer.

Manski, C. F., & Nagin, D. S. (1998). Bounding disagreements about treatment effects: A case study of sentencing and recidivism. *Sociological Methodology*, *28*, 99–137.

Marcus, M., & Sant'Anna, P. H. C. (2021). The role of parallel trends in event study settings: An application to environmental economics. *Journal of Association of Environmental and Resource Economics*, *8*(2), 235–275.

McKie, E. C., Ferguson, M. E., Galbreth, M. R., & Venkataraman, S. (2018). How do consumers choose between multiple product generations and conditions? An empirical study of iPad sales on eBay. *Production and Operations Management*, *27*(8), 1574–1594.

Mejia, J., Urrea, G., & Pedraza-Martinez, A. J. (2019). Operational transparency on crowdfunding platforms: Effect on donations for emergency response. *Production and Operations Management*, *28*(7), 1773–1791.

Melnyk, S. A., Flynn, B. B., & Awaysheh, A. (2018). The best of times and the worst of times: Empirical operations and supply chain management research. *International Journal of Production Research*, *56*(1-2), 164–192.

Miao, W., Deng, Y., Wang, W., Liu, Y., & Tang, C. S. (2021). The effects of surge pricing on driver behavior in the ride-sharing market: Evidence from a quasi-experiment. http://doi.org/10.2139/ssrn.3909409
Mithas, S., Almirall, D., & Krishnan, M. S. (2006). Do CRM systems cause one-to-one marketing effectiveness? *Statistical Science*, *21*(2), 223–233.
Mithas, S., Almirall, D., & Krishnan, M. S. (2014). A potential outcomes approach to assess causality in information systems research. In R. J. Kauffman, & P. P. Tallon (Eds.), *Economics, information systems and electronic commerce research II: Advanced empirical methodologies* (Vol. 2, pp. 63–85). ME Sharpe.
Mithas, S., Chen, Z.-L., Saldanha, T., & De Oliveira Silveira, A. (2022). How will artificial intelligence and Industry 4.0 emerging technologies transform operations management? *Production and Operations Management*, 1–13. https://doi.org/10.1111/poms.13864
Mithas, S., & Krishnan, M. S. (2009). From association to causation via a potential outcomes approach. *Information Systems Research*, *20*(2), 295–313.
Mithas, S., Xue, L., Huang, N., & Burton-Jones, A. (2022). Editor's comments: Causality Meets Diversity in Information Systems Research. *Management Information Systems Quarterly*, *46*(3), iii–xviii.
Morgan, S. L., & Winship, C. (2007). *Counterfactuals and causal inference: Methods and principles for social research*. Cambridge University Press.
Muthulingam, S., Dhanorkar, S., & Corbett, C. J. (2021). Does water scarcity affect environmental performance? Evidence from manufacturing facilities in Texas. *Management Science*, *68*(4), 2377–3174.
Park, S., Rabinovich, E., Tang, C. T. S., & Yin, R. (2020). The impact of disclosing inventory-scarcity messages on sales in online retailing. *Journal of Operations Management*, *66*(5), 534–552.
Park, Y. K., & Mithas, S. (2020). Organized complexity of digital business strategy: A configurational perspective. *MIS Quarterly*, *44*(1), 85–127.
Peng, D. X., & Lai, F. (2011). Using partial least squares in operations management research: A practical guideline and summary of past research. *Journal of Operations Management*, *30*(6), 467–480.
Pratt, J. W., & Schlaifer, R. (1988). On the interpretation and observation of laws. *Journal of Econometrics*, *39*(1-2), 23–52.
Rai, A., Patnayakuni, R., & Patnayakuni, N. (1997). Technology investment and business performance. *Communications of the ACM*, *40*(7), 89–97.
Roberts, M. R., & Whited, T. M. (2013). Endogeneity in empirical corporate finance. In G. M. Constantinides, M. Harris, & R. M. Stulz (Eds.), *Handbook of the economics of finance* (Vol. 2, Part A, pp. 493–572). Elsevier. https://doi.org/10.1016/B978-0-44-453594-8.00007-0
Rosenbaum, P. R. (1999). Choice as an alternative to control in observational studies. *Statistical Science*, *14*(3), 259–304.
Rosenbaum, P. R. (2019). *Design of observational studies* (2nd ed.). Springer.
Rosenbaum, P. R. (2002). *Observational studies* (2nd ed.). Springer.
Rosenbaum, P. R., & Rubin, D. B. (1983a). Assessing sensitivity to an unobserved binary covariate in an observational study with binary outcome. *Journal of Royal Statistical Society Series B*, *45*(2), 212–218.
Rosenbaum, P. R., & Rubin, D. B. (1983b). The central role of the propensity score in observational studies for causal effects. *Biometrika*, *70*(1), 41–55.
Roth, A., Tucker, A. L., Venkataraman, S., & Chilingerian, J. (2019). Being on the productivity frontier: Identifying "triple aim performance" hospitals. *Production and Operations Management*, *28*(9), 2165–2183.
Roth, A., & Rosenzweig, E. (2020). Advancing empirical science in operations management research: A clarion call to action. *Manufacturing & Service Operations Management*, *22*(1), 179–190.
Rubin, D. B. (1997). Estimating causal effects from large data sets using propensity scores. *Annals of Internal Medicine*, *127*(8 Pt 2), 757–763.
Saldanha, T., Lee, D., & Mithas, S. (2020). Aligning information technology and business: The differential effects of alignment during investment planning, delivery, and change. *Information Systems Research*, *31*(4), 1260–1281.
Shadish, W. R., Cook, T. D., & Campbell, D. T. (2002). *Experimental and quasi-experimental designs for generalized causal inference*. Wadsworth Cengage Learning.
Shah, R., & Goldstein, S. M. (2006). Use of structural equation modeling in operations management research: Looking back and forward. *Journal of Operations Management*, *24*(2), 148–169.
Sobel, M. E. (1995). Causal inference in the social and behavioral sciences. In G. A. Arminger, C. C. Clogg, & M. E. Sobel (Eds.), *Handbook of statistical modeling for the social and behavioral sciences.* Plenum Press.
Song, P., Wang, Q., Liu, H., & Li, Q. (2020). The value of buy-online-and-pickup-in-store in omni-channel: Evidence from customer usage data. *Production and Operations Management*, *29*(4), 995–1010.
Stock, J., Yogo, M., & Wright, J. (2002). A survey of weak instruments and weak identification in generalized method of moments. *Journal of Business and Economic Statistics*, *20*, 518–529.
Sun, L., & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogenous treatment effects. *Journal of Econometrics*, *225*(2), 175–199.
Tang, C. S., & Veelenturf, L. P. (2019). The strategic role of logistics in the industry 4.0 era. *Transportation Research Part E*, *129*, 1–11.
Tang, Q., Lin, M., & Kim, Y. (2021). Inter-retailer channel competition: Empirical analyses of store entry effects on online purchases. *Production and Operations Management*, *30*(8), 2547–2563.
Tang, Q., & Whinston, A. B. (2020). Do reputational sanctions deter negligence in information security management? A field quasi-experiment. *Production and Operations Management*, *29*(2), 410–427.
Terwiesch, C., Olivares, M., Staats, B. R., & Gaur, V. (2020). A review of empirical operations management over the last two decades. *Manufacturing & Service Operations Management*, *22*(4), 645–867.
The Economist. (2016, November 26). Economists are prone to fads, and the latest is machine learning. *The Economist*. https://www.economist.com/finance-and-economics/2016/11/24/economists-are-prone-to-fads-and-the-latest-is-machine-learning
Wandersee, J. H. (1990). Guest editorial: Imprecise thinking: A Baconian checklist. *Journal of Research in Science Teaching*, *27*(2), 95–96.
Wang, G., Zheng, R., & Dai, T. (2022). Does transportation mean transplantation? Impact of new airline routes on sharing of cadaveric kidneys. *Management Science*, *68*(5), 3660–3679.
Winship, C., & Morgan, S. L. (1999). The estimation of causal effects from observational data. *Annual Review of Sociology*, *25*, 659–706.
Xie, Y. (2011). Causal inference and heterogeneity bias in social science. *Information Knowledge Systems Management*, *10*(1-4), 279–289.
Xue, L., Song, P., Rai, A., Zhang, C., & Zhao, X. (2019). Implications of application programming interfaces for third-party new app development and copycatting. *Production and Operations Management*, *28*(8), 1887–1902.
Zhang, X., & Yao, Y. (2020). How much is too much? The effect of offline call intensity on online purchase of digital services. *Production and Operations Management*, *29*(3), 509–525.
Zhao, W., Liu, Q. B., Guo, X., Wu, T., & Kumar, S. (2021). Quid pro quo in online medical consultation? Investigating the effects of small monetary gifts from patients. *Production and Operations Management*, *31*(4), 1698–1718.
Zhao, X., Lynch, J. G. Jr., & Chen, Q. (2010). Reconsidering Baron and Kenny: Myths and truths about mediation analysis. *Journal of Consumer Research*, *37*(2), 197–206.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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